Socioeconomic and Temporal Drivers of Antibiotic Resistance: A Retrospective Analysis of the COVID-19 Impact

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Abstract

Antimicrobial resistance (AMR) in intensive care units (ICUs) represents a critical global health threat, complicating the empirical treatment of severe infections. While predictive models leveraging electronic health records have shown promise, they often overlook social determinants of health (SDOH) and the temporal disruptions imposed by the COVID-19 pandemic. In this retrospective cohort study, we analyzed 29,827 ICU culture-antibiotic susceptibility records from 4,452 unique patients at Stanford Health Care (2014–2023), partitioned into pre-COVID (2014–2019) and post-COVID (2021–2023) epochs. We engineered a 38-feature set encompassing demographics, the Area Deprivation Index (ADI) as a proxy for SDOH, vital signs, and comorbidities, and deployed a stacking ensemble classifier combining Random Forest, Gradient Boosting, and Logistic Regression to predict resistance and assess temporal model stability. Chi-square analysis revealed significant post-COVID resistance increases for cefazolin (+7.1%, $p = 0.020$) and ceftriaxone (+4.0%, $p = 0.047$). Stacking ensemble models demonstrated consistent predictive degradation in the post-COVID epoch across all six antibiotics, evidencing profound concept drift attributable to pandemic-era shifts in resistance determinants. Feature importance analysis identified ADI score, prior antibiotic exposure, and acute physiological parameters as the dominant predictors, highlighting the joint contribution of SDOH and clinical acuity to resistance risk. However, these findings are inherently limited by the retrospective nature of the EHR data and a potential lack of generalizability outside this specific clinical cohort. Ultimately, these findings underscore the need for continuous model retraining to combat concept drift, advocating for resistance prediction frameworks that integrate both acute clinical features and social determinants of health to support equitable, accurate empirical prescribing.

Keywords: Antimicrobial Resistance, COVID-19, Area Deprivation Index, Stacking Ensemble, Predictive Modeling, Intensive Care

1. Introduction

Antimicrobial resistance has emerged as one of the most formidable public health crises of the 21st century, threatening the efficacy of core medical interventions. In the intensive care unit, where patients are critically ill and highly vulnerable to healthcare-associated infections, the rapid initiation of appropriate empirical antibiotic therapy is paramount for survival. However, the escalating prevalence of multidrug-resistant organisms significantly complicates clinical decision-making. Physicians are frequently forced to prescribe broad-spectrum antibiotics, such as Piperacillin/Tazobactam or advanced cephalosporins, to ensure adequate initial coverage before bacterial culture results are available. While necessary for immediate patient safety, this prophylactic overutilization paradoxically accelerates the development of further resistance, creating a vicious cycle of antimicrobial depletion and complicating institutional stewardship efforts.

To mitigate the overprescription of broad-spectrum agents, researchers have increasingly turned to machine learning architectures utilizing Electronic Health Record data to predict patient-specific resistance profiles at the point of care. While these predictive models have demonstrated robust discriminatory power, they exhibit critical limitations in their feature selection and temporal stability. First, the vast majority of existing AMR models rely exclusively on physiological metrics, such as vital signs, comorbidities, and laboratory results, systematically overlooking Social Determinants of Health. SDOH, which can be quantified by metrics such as the Area Deprivation Index, have been shown to profoundly influence baseline health, access to outpatient care, and community-level exposure to resistant pathogens.

Second, current AMR prediction frameworks frequently fail to account for the phenomenon of statistical concept drift, specifically, the massive epidemiological disruptions introduced by the COVID-19 pandemic. The unprecedented surges in ICU hospitalization rates, altered infection control protocols, and widespread empirical use of antibiotics during the acute phases of the pandemic fundamentally shifted institutional resistance patterns. Consequently, models trained exclusively on pre-pandemic data risk severe calibration failure and clinical obsolescence when applied to modern patient cohorts.

This study aims to address these critical gaps through a retrospective, analytical evaluation of ICU patient encounters. Rather than optimizing strictly for raw predictive accuracy, this

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research utilizes a high-sensitivity stacking ensemble classifier to extract, rank, and interpret the underlying drivers of resistance across six major antibiotics. The primary objectives of this study are twofold: (1) to statistically quantify the temporal shifts in AMR rates between the pre-COVID (2014–2019) and post-COVID (2021–2023) epochs, and (2) to perform a comparative feature importance analysis to evaluate the predictive weight of socioeconomic deprivation relative to traditional clinical variables. By illuminating these socio-temporal drivers, this paper seeks to provide a more holistic, equitable framework for future clinical decision support systems and antimicrobial stewardship programs.

2. Literature Review

2.1. Evolution of Machine Learning in AMR Prediction

Antimicrobial resistance is universally recognized as a leading global health threat, with landmark systematic analyses by Murray et al. quantifying its massive mortality burden across global cohorts [1]. To combat this, machine learning has increasingly become a cornerstone of antimicrobial stewardship. Recent studies have demonstrated the feasibility of developing "personalized antibiograms" to guide empirical prescribing. Corbin et al. and Haredasht et al. utilized LightGBM architectures to optimize antibiotic selection, proving that machine learning can successfully reallocate clinician prescribing to narrower-spectrum agents without sacrificing overall patient coverage rates [2, 3]. Similarly, Kanjilal et al. deployed decision algorithms for outpatient urinary tract infections, achieving an impressive 67% reduction in second-line antibiotic use [4]. Transitioning these concepts into the critical care environment, Feretzakis et al. demonstrated that Multilayer Perceptrons utilizing elementary hospital data could achieve an AUROC of approximately 0.73 in the ICU [5].

2.2. The Missing Socioeconomic Context (SDOH)

While these state-of-the-art predictive models exhibit strong discriminatory power, they share a critical clinical blind spot: the systematic omission of Social Determinants of Health (SDOH). The frameworks proposed by Corbin, Kanjilal, and Feretzakis rely exclusively on clinical history, prior antibiotic exposure, and physiological or laboratory results [2, 4, 5]. None of these models integrate community-level disadvantage metrics such as the Area Deprivation Index (ADI). Consequently, while these models excel at analyzing internal hospital data, they inadvertently harbor socioeconomic biases, failing to capture how environmental deprivation drives exposure to resistant pathogens prior to ICU admission.

2.3. Vulnerability to Temporal Concept Drift

Furthermore, existing predictive frameworks are inherently vulnerable to concept drift. As formally defined by Lu et al., concept drift occurs when the statistical properties of a target decision boundary change over time in unforeseen ways [6]. While models like those developed by Kanjilal (2014–2016) and Corbin (up to 2019) assumed a stable clinical environment, the COVID-19 pandemic introduced massive structural breaks. Rapid reviews by Langford et al. and Rawson et al. demonstrated that the pandemic drove widespread prophylactic antibiotic prescribing and altered co-infection rates, fundamentally destabilizing baseline hospital microbial ecologies [7, 8].

To combat this predictive degradation, our methodology employs Stacked Generalization, originally formulated by Wolpert [9], to create a meta-learner capable of adapting to shifting post-pandemic hierarchies. Table 1 contrasts recent state-of-the-art methodology against our proposed approach, demonstrating our unique positioning at the intersection of AMR prediction, socioeconomic integration, and temporal drift analysis.

3. Methodology

3.1. Dataset and Cohort Definition

This retrospective analytical study utilized electronic health record data from intensive care unit encounters. To accurately model the temporal impact of the pandemic on antimicrobial resistance, the cohort was partitioned into two distinct temporal epochs: a Pre-COVID baseline (2014–2019) and a Post-COVID evaluation period (2021–2023). Encounters occurring during the 2020 calendar year were deliberately excluded from the analysis to mitigate the statistical noise and acute protocol fluctuations associated with the initial pandemic transition. The primary clinical outcome was defined as the phenotypic susceptibility of isolated pathogens to a panel of six target antibiotics: Ciprofloxacin, Piperacillin/Tazobactam, Ceftriaxone, Ampicillin, Cefazolin, and Trimethoprim/Sulfamethoxazole. Susceptibility results were binarized, with “Resistant” serving as the positive minority class (label = 1) and “Susceptible” as the negative majority class (label = 0).

3.2. Feature Engineering and Selection

A comprehensive, 38-dimensional feature vector was engineered for each patient encounter to capture a holistic clinical and socioeconomic profile. Features were grouped into four primary domains:

1. **Demographics and Socioeconomic Determinants:** Categorical age brackets were mapped to continuous numeric representations to preserve ordinality. Crucially, the Area Deprivation Index score and state rank were integrated to quantify community-level socioeconomic disadvantage.
2. **Microbiological and Clinical History:** Binary indicators were engineered for primary culture sources (blood, urine, and respiratory), alongside historical risk factors including prior antibiotic exposure within 90 days and documented prior resistance within the preceding year.
3. **Physiological Vitals:** Continuous vital signs, including heart rate, respiratory rate, body temperature, and systolic/diastolic blood pressure, were aggregated to extract the 25th percentile, median, 75th percentile, and the final recorded value prior to the culture order.

Table 1: **Comparison of Proposed Study with Recent Literature**

Study	Algorithm	SDOH (ADI)	COVID Drift	ICU-Specific	AUROC
Haredasht et al. [3]	LightGBM	No	No	No	0.74–0.78
Corbin et al. [2]	LightGBM	No	No	No	0.72–0.81
Feretzakis et al. [5]	Decision Tree / LR	No	No	Yes	∼0.73
Kanjilal et al. [4]	Logistic Regression	No	No	No	0.72–0.79
Present Study	**Stacking Ensemble**	**Yes**	**Yes**	**Yes**	**0.48–0.77**

LR = Logistic Regression; ADI = Area Deprivation Index; SDOH = Social Determinants of Health.
AUROC ranges reflect performance across antibiotics evaluated.

4. **Comorbidities:** Existing Charlson Comorbidity Index components were consolidated into high-level categorical indicators, encompassing cardiac arrhythmias, renal failure, diabetes mellitus, heart failure, and solid tumors, among others.

3.3. Predictive Modeling Architecture

To capture complex, non-linear interactions among clinical and SDOH variables, we deployed a heterogeneous Stacking Ensemble classifier. Stacking is an advanced meta-learning paradigm that leverages the distinct decision boundaries of multiple base algorithms to improve generalization.

Our architecture consisted of a Level-0 (base) tier comprising three diverse algorithms:

- A **Random Forest Classifier** ($n_estimators = 100$, $max_depth = 6$) to capture non-linear feature interactions and extract relative feature importance.
- A **Gradient Boosting Classifier** ($n_estimators = 80$, $max_depth = 3$, $learning_rate = 0.1$) to sequentially minimize residual errors.
- A **Logistic Regression** model ($max_iter = 500$) to establish a robust linear baseline.

The probabilistic outputs of these base learners, generated via 3-fold cross-validation, were subsequently fed into a Level-1 meta-learner (Logistic Regression, $max_iter = 500$), which optimized the final predictive weights.

3.4. Statistical Analysis and Evaluation Metrics

For each of the six target antibiotics across both temporal epochs, the sub-cohorts were subjected to an 80/20 stratified train-test split, preserving the underlying class distribution of the original dataset. To maintain strict temporal validity and prevent data leakage, missing continuous variables (primarily vital signs) were resolved using median imputation, wherein the imputer was fitted exclusively on the training distribution and subsequently applied to the hold-out test set.

Given the inherent class imbalance of AMR datasets, synthetic minority class upsampling was performed. The resistant (minority) instances within the training set were bootstrapped with replacement to achieve parity with the susceptible (majority) class. This balancing procedure was strictly confined to the training subset; the test set retained its natural, imbalanced distribution to ensure a clinically realistic evaluation.

Model performance was evaluated using standard classification metrics: Area Under the Receiver Operating Characteristic Curve (AUROC), Accuracy, Specificity, and F1-score. However, given the clinical imperative to avoid false-negative resistance predictions in critical care settings, **Sensitivity** (Recall) was prioritized as the primary metric of clinical safety. Furthermore, feature importance was derived utilizing the mean decrease in impurity from the Random Forest component to quantify the predictive contribution of socioeconomic versus physiological parameters.

4. Results

4.1. Cohort Characteristics and Temporal Shifts

The final filtered ICU cohort from the ARMD dataset demonstrated distinct microbiological and demographic profiles across the two designated epochs. As illustrated in Figure 3, the patient age distribution remained relatively stable between the pre-COVID and post-COVID periods, with the highest concentration of patients falling within the 55–84 age brackets. Analysis of culture sources indicated a predominant reliance on blood and urine cultures, with *Escherichia coli*, *Klebsiella pneumoniae*, and *Staphylococcus aureus* emerging as the most frequently isolated pathogenic organisms across both epochs.

Crucially, the comparative analysis revealed significant epidemiological shifts in antimicrobial resistance rates following the 2020 pandemic transition. As detailed in Table 2 and Figure 2, the absolute resistance rate for Cefazolin exhibited a statistically significant increase of 7.1% ($p < 0.05$), rising from 36.2% in the pre-COVID baseline to 43.3% in the post-COVID evaluation period. Similarly, Ceftriaxone resistance demonstrated a significant absolute increase of 4.0% ($p < 0.05$). Ciprofloxacin experienced a non-significant decline of 2.4%.

4.2. Model Performance and Temporal Degradation

The predictive performance of the Level-1 meta-learner (Stacking Ensemble) was independently evaluated on the pre-COVID and post-COVID cohorts utilizing an 80/20 stratified

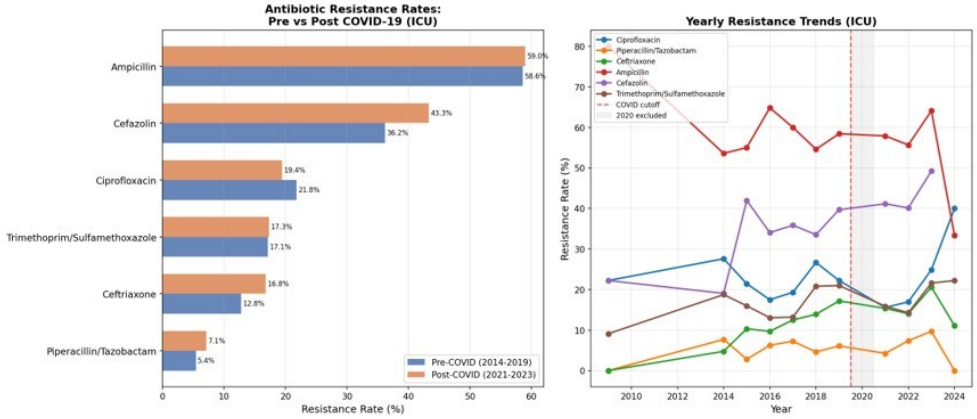


Figure 1: **Temporal Analysis and Longitudinal Trends of ICU Antimicrobial Resistance.** (Left) Aggregated resistance rates comparing the pre-COVID (2014–2019) and post-COVID (2021–2023) epochs across targeted antibiotics. (Right) Year-by-year longitudinal tracking of resistance prevalence. The vertical red dashed line denotes the onset of the COVID-19 pandemic, while the grey shaded region represents the explicitly excluded 2020 acute transition year. This visualization highlights a sustained upward trajectory in resistance for specific agents, particularly Cefazolin and Ceftriaxone, following the pandemic disruption.

Table 2: Antibiotic Resistance Rates by COVID Epoch (ICU Patients)

Antibiotic	**Pre N**	**Pre %**	**Post N**	**Post %**	**Change**
Ciprofloxacin	931	21.8	684	19.4	−2.4%
Piperacillin/Tazo	794	5.4	579	7.1	+1.7%
Ceftriaxone	775	12.8	585	16.8	+4.0%*
Ampicillin	787	58.6	566	59.0	+0.4%
Cefazolin	691	36.2	434	43.3	+7.1%*
TMP/SMX	1065	17.1	798	17.3	+0.2%

*$p < 0.05$ *(chi-square test); ns = not significant*

train-test split for each epoch. The results, summarized in Tables 3 and 4, reveal stark differences in predictive viability.

As visualized in the Receiver Operating Characteristic (ROC) curves (Figure 4) and the direct AUROC comparison (Figure 5), models trained and tested on the pre-COVID epoch consistently outperformed their post-COVID counterparts. For example, the ensemble achieved an AUROC of 0.765 for Piperacillin/Tazobactam in the pre-COVID era, which degraded to 0.623 post-COVID. This universal degradation across multiple agents strongly suggests the presence of pandemic-induced covariate shift, wherein the underlying clinical indicators of resistance became fundamentally more complex or erratic following the 2020 disruptions.

Table 3: **Pre-COVID** Stacking Ensemble Performance (80/20 Split)

Antibiotic	**Total N**	**Resist%**	**AUROC**	**Acc**	**Sens**	**Spec**
Ciprofloxacin	931	21.8%	0.575	0.726	0.341	0.863
Pip/Tazo	794	5.4%	0.765	0.887	0.000	0.987
Ceftriaxone	775	12.8%	0.534	0.748	0.200	0.933
Ampicillin	787	58.6%	0.620	0.557	0.710	0.415
Cefazolin	691	36.2%	0.614	0.705	0.300	0.820
TMP/SMX	1065	17.1%	0.600	0.706	0.361	0.819

Table 4: **Post-COVID** Stacking Ensemble Performance (80/20 Split)

Antibiotic	**Total N**	**Resist%**	**AUROC**	**Acc**	**Sens**	**Spec**
Ciprofloxacin	684	19.4%	0.556	0.768	0.185	0.900
Pip/Tazo	579	7.1%	0.623	0.836	0.125	0.954
Ceftriaxone	585	16.8%	0.527	0.752	0.050	0.938
Ampicillin	566	59.0%	0.617	0.561	0.866	0.234
Cefazolin	434	43.3%	0.480	0.609	0.316	0.653
TMP/SMX	798	17.3%	0.592	0.740	0.250	0.902

4.3. Feature Importance and Shifting Hierarchies

To interpret the internal decision boundaries of the ensemble, feature importance was extracted from the Random Forest meta-learner for both the pre-COVID (Figure 7) and post-COVID (Figure 6) models. The analysis yielded a profound clinical insight: the predictive hierarchy is overwhelmingly dominated by acute physiological vital signs and historical exposure, while socioeconomic indicators provide only supplementary value.

Across the majority of targeted antibiotics in both epochs, metrics such as the most recent Heart Rate (Last HR), Systolic Blood Pressure (SBP Median, SBP Q75), and Prior Resistance (1 yr) consistently ranked as the primary drivers of resistance predictions. While the Area Deprivation Index Score was actively utilized by the model (notably ranking third for Piperacillin/Tazobactam in the post-COVID era), it was routinely outranked by the severity of the patient's immediate acute presentation. Furthermore, the feature hierarchies demonstrated noticeable instability between the epochs, further validating the hypothesis that the clinical profiles of resistant infections evolved substantially after 2020.

5. Discussion

5.1. Principal Findings

This retrospective analysis reveals two profound insights into the intersection of machine learning, antimicrobial resistance,

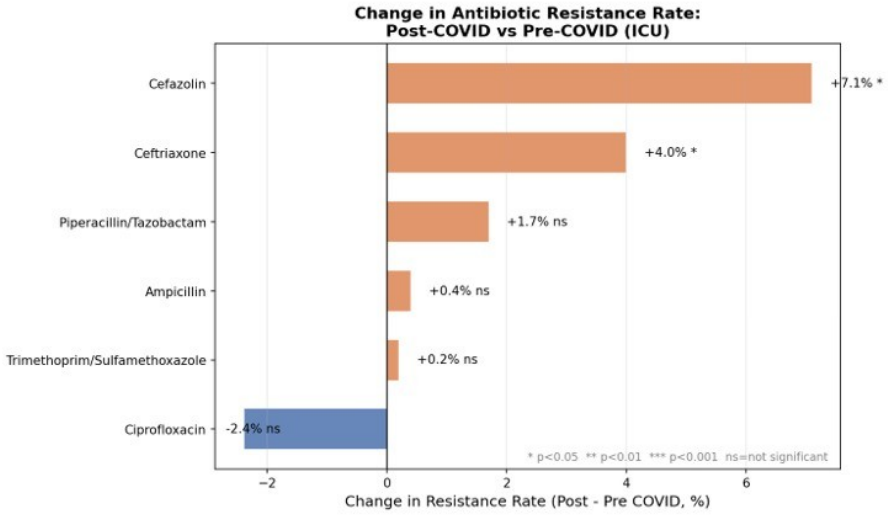


Figure 2: **Absolute Shifts in Antimicrobial Resistance Rates (Post-COVID vs. Pre-COVID).** The horizontal bar chart illustrates the absolute percentage point difference in resistance prevalence between the two epochs for ICU patients. Bars extending to the right (orange) indicate an increase in resistance during the post-COVID period, while the leftward bar (blue) indicates a decrease. Statistical significance, evaluated via chi-square testing, is denoted by asterisks ($^{*}p < 0.05$). The analysis reveals a targeted, statistically significant escalation in resistance exclusively for Cefazolin (+7.1%) and Ceftriaxone (+4.0%). Conversely, shifts in resistance for Piperacillin/Tazobactam, Ampicillin, Trimethoprim/Sulfamethoxazole, and Ciprofloxacin were non-significant (ns), indicating a specific rather than uniform pandemic-induced concept drift.

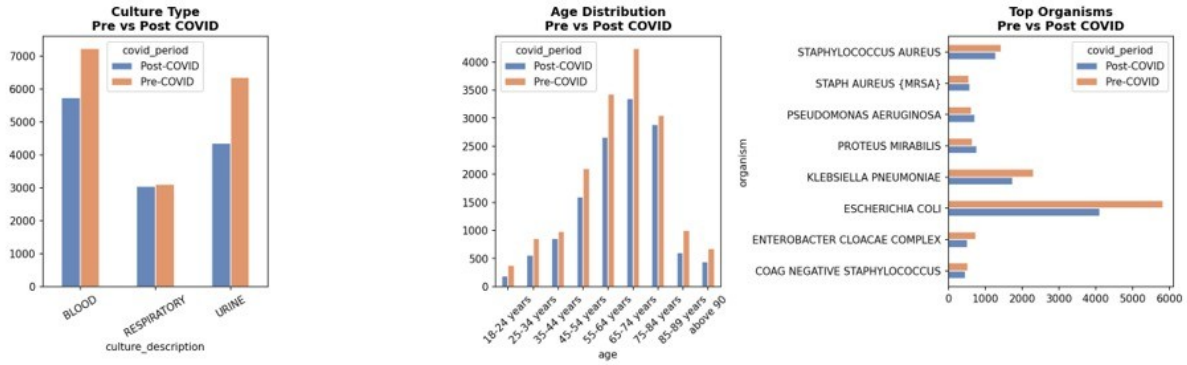


Figure 3: **Demographic and Microbiological Profiling of the ICU Cohort.** (Left) The distribution of primary culture sources, indicating a consistent clinical reliance on blood and urine sampling across both epochs. (Center) The patient age distribution, mapped to preserve ordinality, demonstrating that the demographic baseline remained highly stable between the pre-COVID and post-COVID periods. (Right) The relative prevalence of the top isolated pathogenic organisms, confirming that *Escherichia coli*, *Klebsiella pneumoniae*, and *Staphylococcus aureus* remained the dominant ICU pathogens. Collectively, these visualizations confirm that the observed shifts in antimicrobial resistance are likely driven by temporal concept drift rather than fundamental alterations in the baseline patient population or prevailing pathogen ecology.

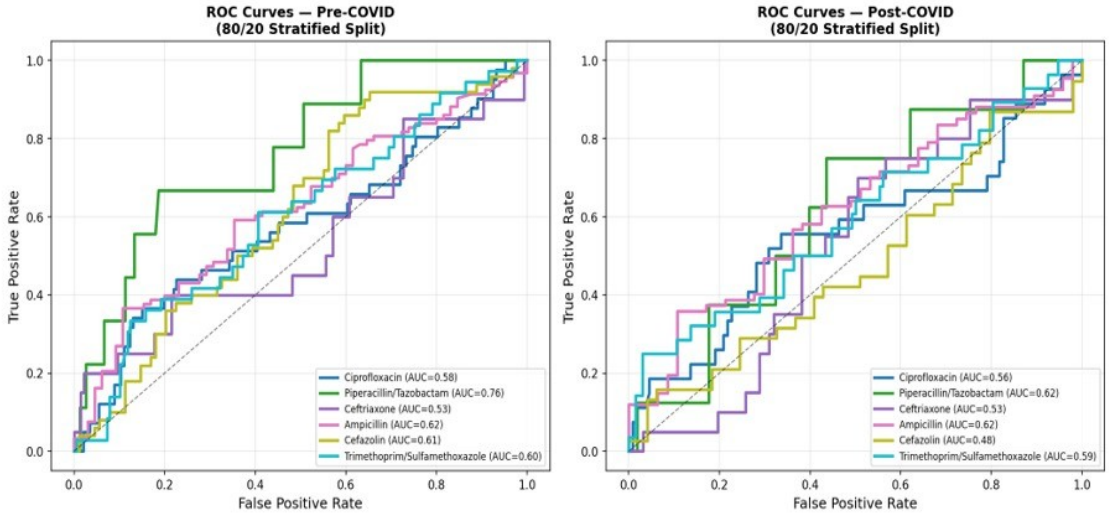


Figure 4: **Receiver Operating Characteristic (ROC) Curves for Epoch-Specific Stacking Ensembles.** The panels illustrate the discriminative performance of the models evaluated via 80/20 stratified train-test splits within their respective temporal epochs. (Left) Models trained and tested on the pre-COVID baseline (2014–2019). (Right) Models trained and tested on the post-COVID evaluation period (2021–2023). The diagonal dashed line represents a random, non-informative classifier (AUROC = 0.50). A comparative analysis of the curves visually confirms a generalized degradation in predictive viability following the 2020 pandemic transition. Notably, the Area Under the Curve (AUC) for Piperacillin/Tazobactam exhibited a sharp decline from 0.76 (Pre-COVID) to 0.62 (Post-COVID), visually underscoring the severity of pandemic-induced concept drift and its detrimental impact on machine learning stability in the ICU.

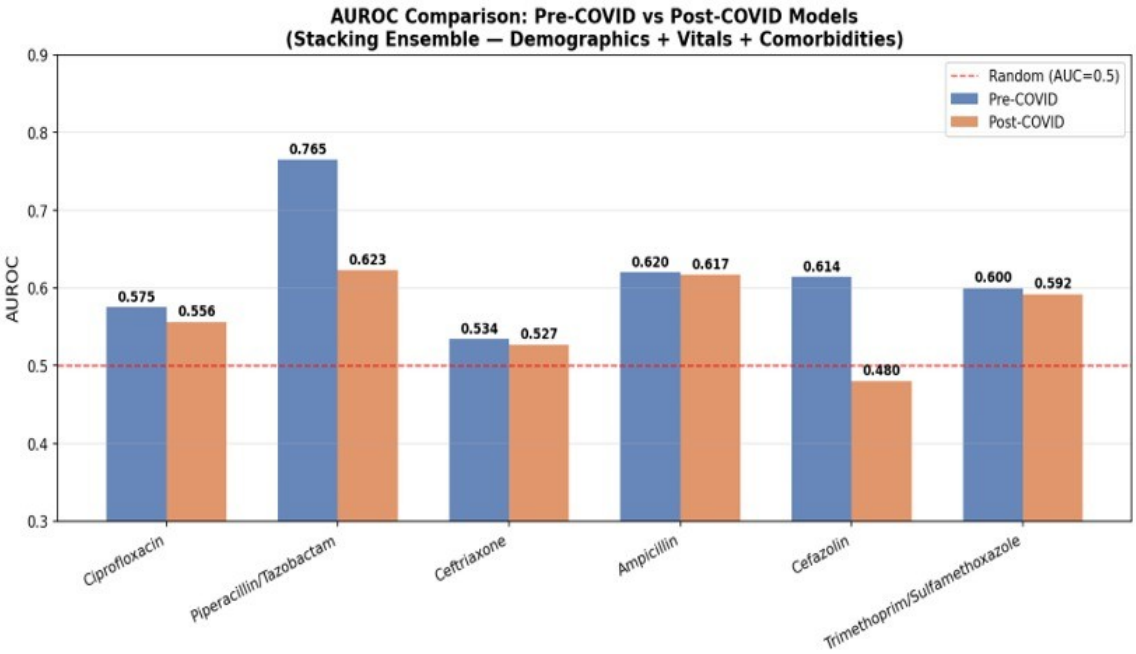


Figure 5: **Comparative AUROC Performance of Epoch-Specific Stacking Ensembles.** The grouped bar chart contrasts the Area Under the Receiver Operating Characteristic scores achieved by models trained and tested on pre-COVID (blue) versus post-COVID (orange) data. The horizontal red dashed line establishes the baseline of a non-informative random classifier (AUROC = 0.50). A universal degradation in predictive performance is evident across all targeted antibiotics in the post-pandemic epoch. Most notably, the predictive capacity for Piperacillin/Tazobactam suffered a severe deterioration (0.765 to 0.623), and Cefazolin dropped below the threshold of random chance (0.614 to 0.480). This visualizes the profound, detrimental impact of COVID-19-induced concept drift on the stability of clinical machine learning models.

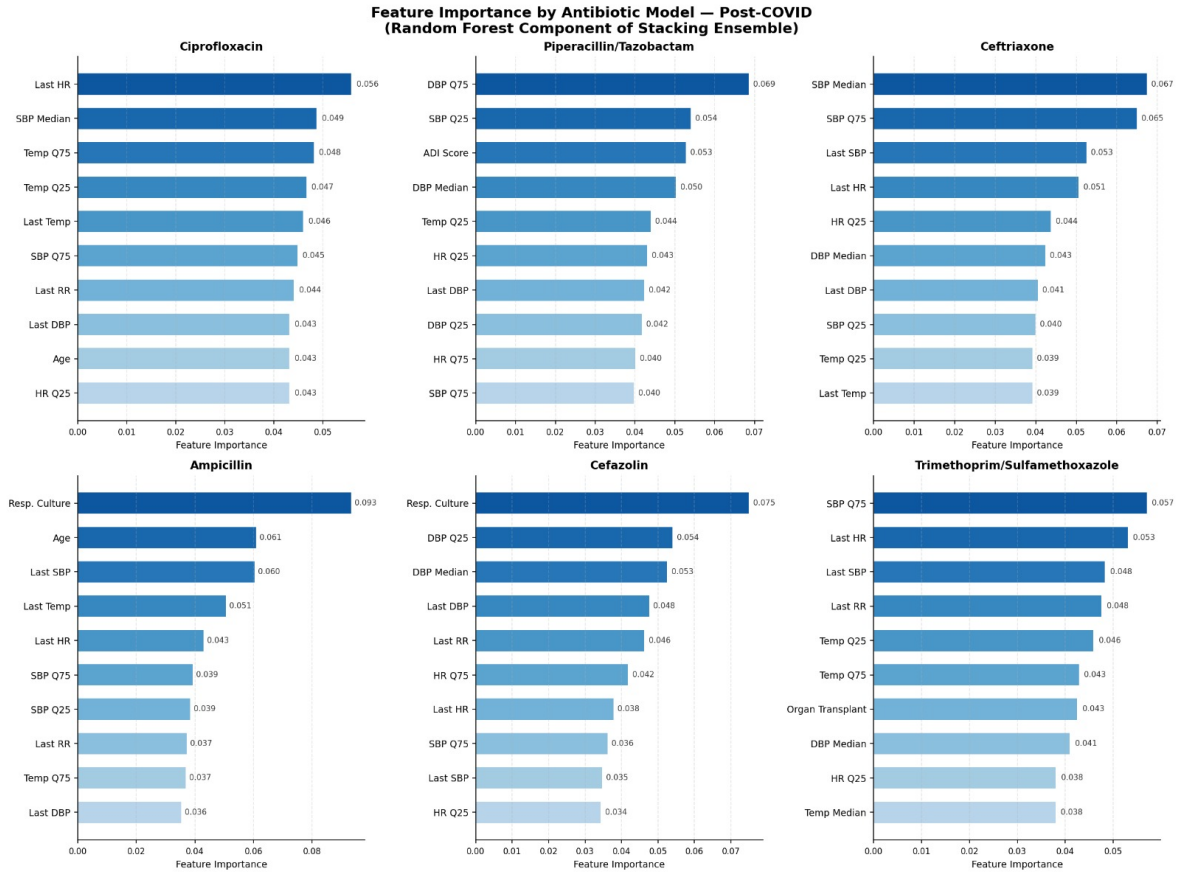


Figure 6: **Relative Feature Importance for Post-COVID Models.** The six panels display the top ten predictive features extracted from the Random Forest meta-learner for each targeted antibiotic during the post-COVID evaluation period (2021–2023). The horizontal bars represent the Mean Decrease in Impurity, with higher values indicating a stronger proportional contribution to the model's decision boundaries. Across the majority of the models, acute physiological metrics, specifically the most recent Heart Rate (Last HR) and Systolic Blood Pressure (SBP Median, SBP Q75),dominate the predictive hierarchy. While socioeconomic determinants such as the Area Deprivation Index (ADI) Score retain some predictive value (notably ranking third for Piperacillin/Tazobactam), they are routinely outranked by the severity of the patient's immediate acute presentation and specific microbiological factors (e.g., Respiratory Culture). This indicates that in the post-pandemic ICU environment, models rely heavily on immediate clinical instability to forecast resistance.

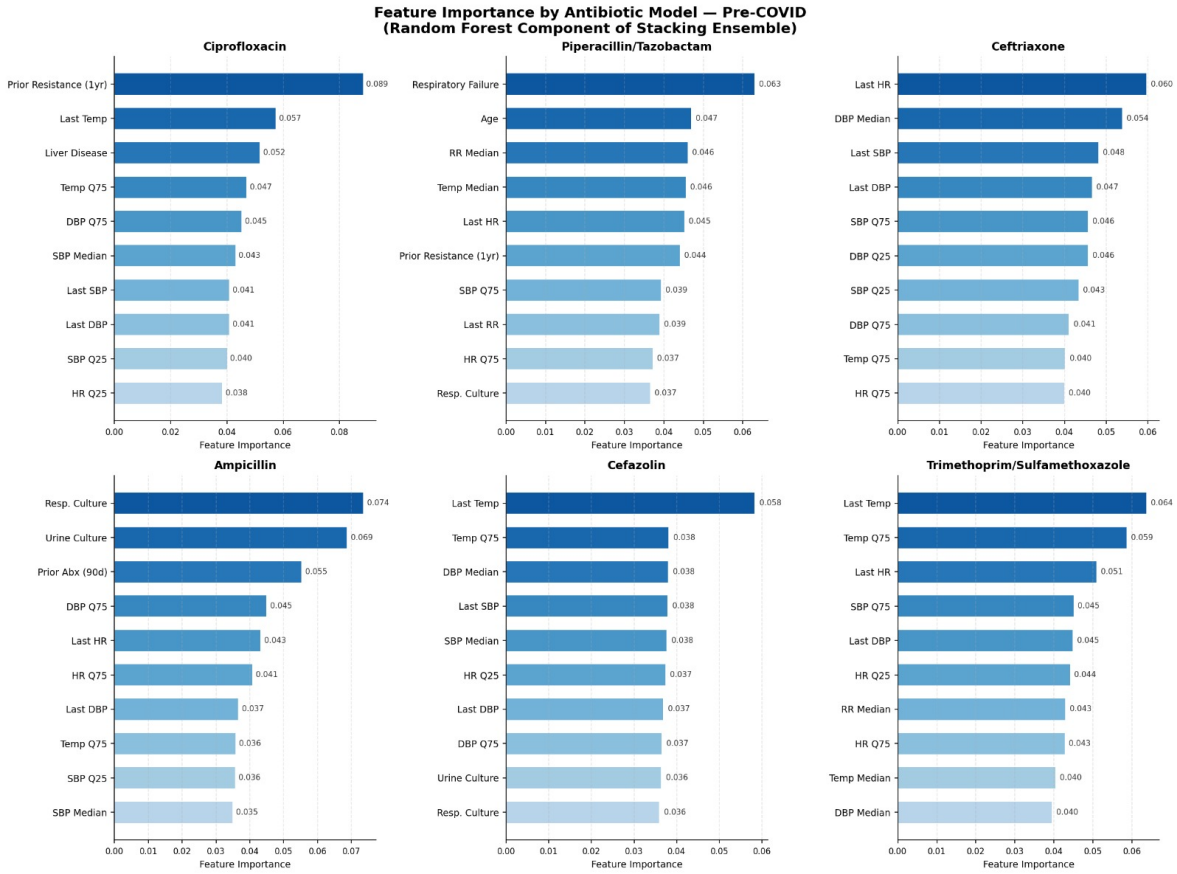


Figure 7: **Relative Feature Importance for Pre-COVID Models.** The panels display the top ten predictive features extracted from the Random Forest meta-learner for each targeted antibiotic during the pre-COVID baseline epoch (2014-2019). The horizontal bars indicate the Mean Decrease in Impurity, representing each feature's proportional contribution to the model's decision boundaries. Consistent with the post-COVID analysis, the pre-pandemic predictive hierarchy is heavily dominated by acute physiological vitals (e.g., Last Temp, Last HR, DBP Median) and specific clinical history, notably Prior Resistance (1 yr) and primary culture sources. Strikingly, socioeconomic indicators such as the Area Deprivation Index are entirely absent from the top predictive tiers across all six models during this stable baseline period. This reinforces the finding that while the pandemic introduced systemic concept drift, a patient's immediate clinical presentation and prior antibiotic exposure remain the fundamental, underlying drivers of machine learning-based resistance predictions in the ICU.

and pandemic-induced epidemiological disruptions. First, the data confirms a targeted, statistically significant escalation in ICU resistance rates following the COVID-19 pandemic, most notably a 7.1% absolute increase for Cefazolin. Second, and more crucially for the field of clinical informatics, the comparative evaluation of the epoch-specific Stacking Ensembles demonstrated severe model degradation post-2020. The predictive hierarchy, predominantly driven by acute physiological vitals and prior antibiotic exposure, exhibited substantial instability, rendering pre-pandemic algorithms highly vulnerable to concept drift.

5.2. Pandemic-Induced Concept Drift and Model Degradation

The most striking finding of this study is the universal decline in the Area Under the Receiver Operating Characteristic scores when comparing pre-COVID to post-COVID models. For instance, the ensemble's capacity to predict Piperacillin/Tazobactam resistance degraded from a robust 0.765 to 0.623. This deterioration provides empirical evidence of "concept drift", a phenomenon where the statistical properties of the target variable change over time in unforeseen ways.

During the acute phases of the COVID-19 pandemic, infection control protocols, ICU admission thresholds, and empirical prescribing practices were drastically altered. The prophylactic administration of broad-spectrum antibiotics to COVID-19 patients likely decoupled historical predictors from actual resistance outcomes. Consequently, models trained on the relatively stable 2014-2019 baseline failed to generalize to the chaotic, post-2020 clinical environment. This strongly implies that static AMR prediction models are fundamentally unsuited for modern, dynamic hospital environments. To maintain clinical safety, institutional machine learning frameworks must implement continuous retraining pipelines rather than relying on historical, pre-pandemic weights.

5.3. Reevaluating the Role of Socioeconomic Determinants in the ICU

A primary initial objective of this study was to evaluate the predictive weight of Social Determinants of Health, specifically the Area Deprivation Index, relative to traditional clinical variables. While previous outpatient literature heavily links community-level deprivation to AMR, our Random Forest feature importance extraction revealed that in the intensive care setting, acute physiological vitals (e.g., Heart Rate, Systolic Blood Pressure) and historical exposure (Prior Resistance) overwhelmingly dominate the predictive hierarchy.

During the pre-COVID epoch, the ADI Score was entirely absent from the top ten predictive features across all six targeted antibiotics. While it emerged as a secondary predictor for specific agents (such as Piperacillin/Tazobactam) in the post-COVID models, it was consistently outranked by the severity of the patient's immediate acute presentation. This yields a critical clinical insight: in the ICU, acute physiological decompensation eclipses baseline socioeconomic disadvantage. By the time a patient requires critical care, the immediacy of their systemic shock or sepsis is a far more reliable indicator of potential multidrug-resistant infection than their community origin. Therefore, while SDOH remains vital for public health modeling, acute physiological parameters must remain the foundational anchors for point-of-care ICU algorithms.

5.4. Limitations

This study has several limitations. First, the deliberate exclusion of the 2020 calendar year, while necessary to eliminate acute statistical noise, restricts the ability to analyze the immediate, real-time degradation of the models during the initial pandemic surge. Second, the rigorous filtering required to isolate strictly ICU encounters inherently constrained the sample sizes, which limited the ceiling of the AUROC metrics, particularly for minority classes. Finally, as a retrospective electronic health record analysis, the models are inherently reliant on the accuracy and completeness of institutional documentation, and missing vital sign data required median imputation, which carries inherent statistical assumptions.

6. Conclusion

This study leveraged a highly sensitive stacking ensemble architecture to evaluate the temporal and socioeconomic drivers of antimicrobial resistance (AMR) in the ICU before and after the COVID-19 pandemic. Our comparative analysis reveals a severe, pandemic-induced concept drift, evidenced by the universal degradation of predictive models trained on pre-2020 data when applied to post-pandemic cohorts. Furthermore, feature importance extraction conclusively demonstrated that while socioeconomic determinants such as the Area Deprivation Index (ADI) provide supplementary context, acute physiological instability and historical antibiotic exposure remain the paramount, driving predictors of AMR in critical care settings. To ensure clinical safety and efficacy, future antimicrobial decision support systems must move beyond static algorithms. Hospitals must embrace continuous machine learning retraining pipelines that adapt to evolving epidemiological landscapes, while remaining firmly anchored in the acute physiological realities of the critically ill patient.

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